

Deterministic Knowledge Conduit

Restructuring AI around human knowledge storage, verified distribution, and individual need.

Tom Klootwijk | UGTOMS concept note v0.1

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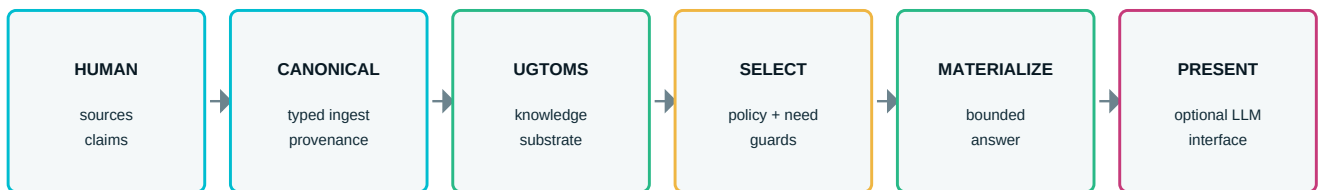
Core proposition

The model should not be the memory, the authority, the policy engine, and the narrator at the same time. UGTOMS separates those roles. Human knowledge becomes typed, versioned, attributable substrate; deterministic gates select what applies; an LLM may remain as an optional interface, never the source of truth.

The restructuring

Position

UGTOMS reorganizes AI from a free-form answer generator into a controlled conduit. It stores claims with meaning and lineage, selects them under explicit rules, materializes only supported outputs, and returns UNKNOWN when the substrate does not justify an answer.



Every authoritative output carries source, version, policy, derivation and an explicit UNKNOWN path.

A technically honest correction

Calling conventional LLMs floating-point brute forcers captures a real frustration but is not technically precise. They are learned statistical functions, commonly executed with floating-point, reduced-precision, or quantized matrix arithmetic. They compress and interpolate patterns; they do not merely enumerate answers. Their central limitation is different: next-token plausibility is not an inherent truth, provenance, permission, or applicability check.

LLM-ONLY ANSWER PATH

- + Prompt mixes fact, policy and style
- + Model predicts plausible tokens
- + Confidence may sound stronger than evidence
- + Missing knowledge can become invented detail

UGTOMS-CONDUIT PATH

- + Facts, claims, policy and presentation are typed
- + Deterministic gates select permitted material
- + Derivation and provenance travel with output
- + Missing support returns UNKNOWN or asks a human

The complete change is architectural: separate source memory, claim semantics, deterministic selection, user policy, derivation, presentation, and action authority. A probabilistic model can still be excellent at proposing links, summarizing, translating, and conversing, while being structurally unable to silently promote a guess into an authoritative fact.

Applications centered on individual need

Individualization is a deterministic materialization profile, not an invitation to manipulate the user. The profile should be inspectable, consented, reversible, and limited to what is needed for the task.

Application	UGTOMS role	Individual materialization	Authority boundary
Education	Prerequisites, concepts, examples, exercises, misconceptions, and mastery evidence as typed relations.	Language, reading level, accessibility, prior mastery, pace, and learning goal.	Tutor may adapt explanation; curriculum facts and assessment rules remain versioned.
Healthcare support	Guidelines, observations, contraindications, evidence levels, and care pathways with provenance.	Condition, language, accessibility, consent, and clinician-approved context.	No autonomous diagnosis. Missing or conflicting support escalates to a qualified human.
Law and civic services	Statutes, versions, jurisdiction, eligibility rules, forms, deadlines, and cited decisions.	Jurisdiction, role, language, case facts, permissions, and requested procedure.	Not legal authority; preserve exact source text and effective dates.
Engineering and manufacturing	Requirements, components, tolerances, procedures, hazards, evidence, and standards adapters.	Machine, material, operator qualification, revision, and task stage.	Simulation, conformance, metrology, and human release remain mandatory.
Science	Datasets, methods, assumptions, units, models, claims, counterclaims, and reproducible derivations.	Field, instrument, uncertainty need, compute budget, and desired abstraction.	Model-generated hypotheses remain proposals until evidence closes the claim.
Personal knowledge	A private lifelong graph of sources, decisions, skills, memories, permissions, and unresolved questions.	Current goal, privacy boundary, attention budget, device, and preferred explanation style.	The person owns correction and deletion controls; sensitive context does not leak into shared output.
Institutions	Policies, procedures, ownership, dependencies, exceptions, incident evidence, and change lineage.	Role, clearance, location, competence, and active responsibility.	Access rules and accountability must be explicit and independently auditable.
Agents and automation	Typed goals, capabilities, preconditions, budgets, action routes, receipts, rollback, and approval gates.	User-approved risk, scope, deadline, and resource constraints.	LLM proposes; deterministic policy and external systems authorize and commit.

Like I am five

Instead of asking a storyteller to remember every book and never make anything up, give it a labeled library, a rule card, and a receipt printer. It may explain the right pages in a way that fits you. If the right page is missing, it must say so rather than invent one.

What makes the conduit deterministic

Layer	Required behavior	Failure behavior
Canonical knowledge	Typed identities, claims, units, scope, version, source, rights, confidence class, and contradiction links.	Unsupported input is preserved literally or quarantined; it is not silently normalized into a claim.
Selection	Deterministic query plan over role, task, permissions, policy, time, location, and evidence requirements.	Ambiguity returns alternatives, requests missing context, or yields UNKNOWN.
Derivation	Versioned bounded operators with declared inputs, outputs, numeric domains, tolerances, and proof or trace hooks.	Out-of-domain, unsafe, or non-convergent operations fail closed.
Personalization	An inspectable profile changes ordering, detail, language, accessibility, examples, and pace.	It must not rewrite source facts, hide relevant risk, or infer sensitive traits without authority.
Presentation	Templates or an optional LLM turn selected material into readable language while preserving citations and claim labels.	Generated wording cannot add unsupported facts; validators compare it with the selected substrate.
Action	Capabilities, preconditions, budgets, approvals, receipts, idempotency, and rollback are separate from conversation.	No verified route means no commit. A plausible sentence is never permission to act.
Update and memory	New observations enter as proposed novelty, pass source and compatibility checks, then receive content-addressed lineage.	Conflicts remain visible; history is not rewritten to create false certainty.

Tests stay, but move out of the creative hot path

Tests are not the product, yet they are how a deterministic conduit earns trust. Authoring should stay fast. Conformance runs at publication, decoder, policy, deployment, and high-risk action boundaries. Golden vectors, independent readers, malformed-input rejection, provenance checks, tolerance budgets, and action receipts prevent the substrate from becoming deterministic theater.

Minimum acceptance rule

Two implementations must agree on canonical identity and declared output; every answer must expose its source and transformation route; missing support must remain missing; and personalization must be reversible without changing authoritative knowledge.

Boundaries and first build

What UGTOMS does not solve by determinism alone

Risk	Why it remains	Required response
False source	A deterministic pipeline can reproduce wrong or malicious information perfectly.	Source reputation, contradiction handling, independent evidence, human governance, and revocation.
Ontology bias	Typed categories can exclude people or encode one institution's worldview.	Plural profiles, appeal paths, visible assumptions, community review, and literal residual preservation.
Privacy	Individual need often requires sensitive context.	Local-first profiles, minimum disclosure, purpose limits, encryption, audit, deletion, and consent.
Stale knowledge	Versioned truth can still be out of date.	Effective dates, freshness rules, dependency invalidation, and explicit stale-state output.
Over-automation	A correct description is not necessarily authority to act.	Separate knowledge, recommendation, authorization, execution, receipt, and rollback layers.
Universal-codec overclaim	Different domains retain different semantics and evidence requirements.	Use a common envelope and adapter contract, not one forced byte layout or one universal truth model.

Small first pilot

Step	Deliverable	Observable pass
1	Ingest one bounded technical manual and its revision history into typed claims plus literal source blocks.	Every claim points to exact source, scope, version, and rights.
2	Define three user profiles: novice operator, qualified technician, and auditor.	Same authority; different order, detail, vocabulary, and permitted operations.
3	Run deterministic selection and template materialization before adding an LLM presenter.	Identical inputs produce identical claim sets and hashes.
4	Add the LLM only as translator, explainer, question parser, and proposal generator.	Unsupported additions are rejected; citations survive paraphrase.
5	Add one bounded agent action with preview, approval, idempotency, receipt, and rollback.	No action occurs without a verified route and explicit authority.

The compact thesis

Human knowledge is the durable substrate. UGTOMS is the typed conduit. Deterministic gates decide what applies. Individual profiles decide how it is materialized. LLMs remain valuable as probabilistic interfaces and proposal engines, but they do not get to define truth, permission, or irreversible action.

Corpus anchors: GSP4 deterministic support/compatibility/guard flow; UGTS-GN packed state and verified-event routing; Foundation content-addressed operator cells; KSEED evidence/novelty ledger; current UGTS ECS, visual graph, polar LUT and Bayer pilot. These demonstrate recurring mechanisms, not a completed universal AI product.

Document status: conceptual application note. No claim of AGI, guaranteed truth, regulatory approval, or universal interoperability.